UNIVERSITÀ DI PAVIA

RESEARCH PROJECT

HOW CAN ITALY REALLY REDUCE ITS ENERGY DEPENDENCE?



A comparative analysis of renewables, efficiency, electrification and grid flexibility.

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WHY ENERGY DEPENDENCE MATTERS

Energy dependence is an economic vulnerability. Italy's reliance on imported energy is transmitted across the whole economy:

Households
Energy bills and transport costs

Firms
Production costs and competitiveness

Inflation
Transmission of international price shocks

Public finances
Emergency subsidies and support measures

Energy security
Exposure to geopolitical and supply risks

Key idea: Energy dependence becomes a macroeconomic problem when external shocks are transmitted to domestic prices, firms and households.

RESEARCH QUESTION & MAIN ARGUMENT

RESEARCH QUESTION

How can Italy effectively reduce its energy dependence while balancing energy security, environmental sustainability and economic affordability?

Core message: The objective is not full energy self-sufficiency, but lower systemic vulnerability.

A credible strategy requires a coordinated portfolio:

- **Energy efficiency** — reduces final energy demand
- **Renewables** — increase domestic low-carbon supply
- **Electrification** — replaces gas and oil in final uses
- **Grid flexibility** — makes renewables reliable and scalable
- **Biomethane & hydrogen** — targeted solutions for hard-to-electrify sectors

ENERGY SECURITY: BEYOND IMPORT DEPENDENCE

Traditional view
Security = sufficient and reliable energy supply.

Broader view
Energy security is multidimensional.

Key message: Import dependence is only one part of energy security.



Availability

Access to sufficient energy resources



Affordability

Stable and reasonable prices



Technology & efficiency

Resilience, innovation and lower energy intensity



Environmental sustainability

Compatibility with decarbonisation



Regulation & governance

Market rules, infrastructure and policy coordination

ENERGY ACCOUNTING:
WHERE DOES DEPENDENCE APPEAR?

Import dependence must be read through the energy balance. Three levels matter — and a policy may reduce dependence in one part of the system but shift it elsewhere.

Level	What it shows	Why it matters
Primary energy	Energy entering the system	Imports of gas, oil, coal, renewables
Final energy	Energy actually consumed by users	Transport, heating, industry, services
Electricity	Transformed energy carrier	Key for electrification and renewables

Example: Heat pumps cut gas in buildings — but the effect depends on how the extra electricity is produced.

Core idea: Reducing dependence in one part of the system can shift it elsewhere.

THE ENERGY TRILEMMA

Three objectives must be balanced — and a policy can improve one dimension while creating pressure on another.

Energy security

Reduce exposure to external supply and price shocks.

Affordability

Keep energy costs sustainable for households, firms and public finances.

Environmental sustainability

Reduce emissions and avoid high-carbon lock-in.

Main trade-off: A policy can improve one dimension while creating pressure on another.

- Renewables cut imports and emissions, but require grid investment.
- Electrification cuts gas and oil use, but raises electricity demand.
- Efficiency cuts demand and costs, but faces behavioural and financial barriers.

ITALY'S STARTING POINT: HIGH IMPORT DEPENDENCE

79.2% → **74.6%**

2022 import dependency

2023 import dependency

- Dependence is mainly concentrated in fossil fuels
- Oil dominates transport
- Natural gas is central for heating, industry and power generation
- Price shocks hit households, firms, inflation and public finances



Key message: Italy's energy dependence is not only a supply issue — it is a macroeconomic vulnerability.

FINAL ENERGY CONSUMPTION: WHERE DEPENDENCE IS GENERATED

Final consumption shows where energy is actually used. Italy's demand is still dominated by fossil fuels.

Oil products

Mainly transport

Natural gas

Buildings, industry, services and power generation

Electricity

Industry, services, households and growing electrification

Renewables & biofuels

Increasing, but still limited in scale

Core idea: Italy cannot reduce dependence only by producing more renewable electricity — a large share of energy use still happens directly through fuels.

SECTORAL EXPOSURE:
DIFFERENT SECTORS, DIFFERENT VULNERABILITIES

Italy's dependence is sector-specific: there is no single problem, but several sectoral vulnerabilities.

Sector	Main dependence channel	Policy implication
Transport	Oil products	EVs, public transport, modal shift, efficiency
Residential buildings	Natural gas for heating	Heat pumps, insulation, renovation
Industry	Gas and electricity prices	Efficiency, electrification, renewable PPAs
Services	Electricity and gas	Efficiency, distributed solar, demand response

Key message: There is no single Italian energy-dependence problem — there are several sectoral vulnerabilities.

ELECTRICITY SYSTEM:
CLEANER, BUT STILL VULNERABLE

311.9
TWh
electricity demand (2024)

51.0
TWh
net electricity imports

- Renewables are expanding
- Natural gas still plays a major role in power generation
- Electrification raises the importance of electricity security



Main trade-off: Electrification reduces direct oil and gas use, but increases pressure on the power system.

METHODOLOGY: HOW WE COMPARE THE OPTIONS

A qualitative multi-criteria assessment. Each option is evaluated against eight criteria and ranked High — Medium — Low.

Criterion	What we evaluate
Import reduction	Does it reduce gas, oil or electricity imports?
Energy security	Does it reduce exposure to shocks and disruptions?
Affordability	Impact on households, firms and public finances?
Implementation time	Short, medium or long term?
Scalability	Can it be deployed at large scale in Italy?
Environmental impact	Does it reduce emissions and externalities?
System integration	Grids, storage, demand response or market redesign?
Regulatory feasibility	Does it depend on permits, incentives or governance?

Ranking system

High

Medium

Low

THREE TYPES OF MEASURES

Not all solutions reduce dependence in the same way. Most options are complements, not substitutes.

Type of measure	Options	Role
Direct	Energy efficiency, renewables, electrification	Reduce demand, increase domestic supply or replace imported fuels
Enabling	Grids, storage, interconnections, demand response	Make renewables and electrification reliable and scalable
Targeted	Biomethane and hydrogen	Address hard-to-electrify sectors

Policy implication: The problem is not choosing one technology — it is coordinating the portfolio.

ENERGY EFFICIENCY:
REDUCING DEMAND FIRST

Efficiency reduces the energy needed for the same service. People and firms do not demand energy itself — they demand services:

- Heating
- Cooling
- Lighting
- Mobility
- Industrial output
- Mechanical power



Why it matters for Italy — efficiency reduces:

- Gas demand in buildings and industry
- Oil demand where transport efficiency is possible
- Electricity demand and peak pressure
- Exposure to international price shocks

Core message: The most secure unit of energy is the one Italy does not need to import, buy or burn.

THE ECONOMICS OF EFFICIENCY: WHY MARKETS UNDERINVEST

Higher upfront cost

↓

Lower future energy bills

Adoption is slowed by market failures:

Policy implication: Match the policy to the failure: labels, audits, standards, loans, targeted grants and carbon pricing.

Barrier	Meaning
Information problems	Consumers may not know future savings
Split incentives	Landlords invest, tenants pay the bills
Capital constraints	Households and SMEs may lack financing
Behavioural biases	Excessive focus on upfront costs
Externalities	Prices ignore emissions and import risk
Rebound effect	Lower cost may increase use of the service

ELECTRIFICATION:
REPLACING GAS & OIL IN FINAL USES

Electrification changes the energy carrier used in final consumption. It can reduce:

- **Natural gas in buildings** *through heat pumps*
- **Oil in transport** *through electric vehicles*
- **Fossil-fuel exposure** *by shifting demand toward electricity*



Core message: Electrification is the bridge between demand-side policy and renewable supply.

Why it matters for Italy

- Transport is still heavily oil-based
- Residential heating is still strongly gas-based
- Electricity can increasingly come from domestic renewables

THE ELECTRIFICATION TRADE-OFF

Electrification does not eliminate energy demand — it transfers part of it to the electricity system.

Benefit	Condition
Heat pumps reduce gas use	Buildings must be efficient and electricity low-carbon
EVs reduce oil use	Charging must be smart and supported by infrastructure
Lower fossil exposure	Extra electricity must come from domestic renewables
Demand becomes flexible	Tariffs, storage and demand response must be available

Main risk: Dependence may shift from gas and oil to electricity generation, imports or grid constraints.

Core message: Electrification works only if it is coordinated with renewables, grids and flexibility.

RENEWABLE ELECTRICITY: DOMESTIC SUPPLY-SIDE PILLAR

Rebalancing the Primary Energy Mix to Reduce Systemic Vulnerability

Solar and wind turn domestic natural resources into electricity. Once installed, they generate power without continuous imports of gas, oil or coal.

Imported Fuels and Price Volatility

- Natural gas accounted for 42% of national electricity production in 2024.
- Relies heavily on imported primary energy sources, exposing the country to international supply risks.
- Marginal gas-fired power plants continue to dictate wholesale market clearing prices.

RENEWABLE ELECTRICITY
Domestic Resource and Low Marginal Cost

- Renewable sources covered a record 51.83% of national electricity production injected in 2024.
- Transforms domestic natural resources into electricity with zero ongoing fuel requirements.
- Shifts the electricity supply curve to the right, displacing high-cost generation via the merit-order effect.

Core message:
Accelerating renewable capacity transforms volatile external fuel risks into stable domestic capital-intensive assets, minimizing the transmission of international market shocks to households and firms.

RENEWABLES ARE NECESSARY, BUT NOT SUFFICIENT

Solar and wind have low marginal costs, but their integration creates systemic costs involving network capacity, permitting barriers, and market design failures.

Issue	Why it matters
Intermittency	Weather-dependent production does not naturally match daily or seasonal demand peaks
Grid congestion	Renewable energy is often produced in the South, far away from northern industrial demand centers.
Curtailment	Clean power output is wasted when transmission networks do not have enough capacity to absorb it
Permitting delays	Low administrative approvals increase investment risk and delay the replacement of imported fuels
Market design	Current electricity market rules do not sufficiently reward flexibility and real-time grid reliability
Social acceptance	New infrastructure developments like large wind farms and power lines often face local opposition

The key challenge for Italy is not only building more renewable capacity, but preparing the energy system to safely absorb, deliver, and value clean power across the country.

FLEXIBILITY:
CAPACITY INTO REAL ENERGY SECURITY

Enabling Infrastructure to Balance Supply and Demand Over Space and Time

Renewable capacity is not enough. Italy must be able to:

- Move clean electricity efficiently across regions from the South to the North.
- Store variable power across different hours to manage daily solar peaks.
- Shift consumer demand away from critical evening peak hours.
- Balance variable solar and wind generation to avoid grid instability.
- Reward flexible resources through updated electricity market designs.

Channel	Role
Grid reinforcement	Moves power from southern production areas to northern demand centers while upgrading distribution lines
Storage	Shifts electricity from surplus generation hours to peak periods using utility batteries and pumped hydro
Interconnections	Shares balancing resources and reserves across European borders to diversify supply conditions
Demand response	Makes consumption flexible by using smart pricing to align consumer habits with renewable availability

Flexibility infrastructure transforms variable renewable capacity into firm, usable energy services, ensuring that domestic clean power can successfully replace imported natural gas and oil.

FLEXIBILITY:
LIMITS, TRADE-OFFS & MARKET DESIGN

Flexibility improves the system, but it does not emerge automatically.

Investment cost
Upfront capital needs for grids, storage, and interconnections are massive and must be recovered through network tariffs.

Distributional effects
Tariff reforms and dynamic pricing can increase energy bills for vulnerable households who cannot easily shift consumption.

Long lead times
Network infrastructure projects involve long planning and permitting cycles, often lagging behind rapid renewable deployment.

Coordination
System integration demands seamless operational alignment between transmission operators, local distributors, regulators, and final consumers.

Regulation
Storage and demand-side resources require clear market rules to ensure fair grid access and stable revenue streams.

Policy implication

Electricity market design must undergo structural reform to move beyond rewarding simple generation volumes. Policy and regulation must establish granular price signals that explicitly compensate long-term resource adequacy, real-time balancing services, geographical location, and overall grid stability, ensuring that flexibility investments become commercially viable assets.

BIOMETHANE & HYDROGEN: TARGETED COMPLEMENTS

Deploying Renewable Gases Exclusively in Hard-to-Electrify Sectors

SCOPE
Targeting specific industrial and transport niches where direct electrification options face severe technical constraints or prohibitive economic costs.

Hard-to-electrify sectors

- High-temperature industrial heat applications.
- Chemical feedstocks and oil refineries.
- Heavy transport and long-haul logistics segments.
- Power system backup and residual grid flexibility.

Biomethane

- **Grid compatibility:** Can partially replace imported natural gas immediately by utilizing the existing gas infrastructure without end-user adaptation costs.
- **Supply constraints:** Total production is strictly limited by the regional availability of sustainable organic waste and agricultural residues.

Hydrogen

- **Production penalty:** It is an energy carrier, not a primary source, requiring massive volumes of renewable electricity and generating high conversion losses.
- **Logistics hurdles:** Low energy density by volume makes compression, dedicated transport pipelines, and long-term storage highly expensive.

Biomethane and hydrogen must be reserved strictly for hard-to-abate sectors. Public policy must avoid wasting these scarce resources in areas where direct electrification and energy efficiency are already cheaper and more efficient.

COMPARATIVE MATRIX: WHICH OPTIONS PERFORM BEST?

Option	Import reduction	Security	Affordability	Scalability	Integration
Energy efficiency	**High**	**High**	**High**	**High**	Positive
Renewable electricity	**High**	**High**	**Med–High**	**High**	Needs flexibility
Heat pumps	**High (gas)**	**High**	**Medium**	**Med–High**	Needs grid
Electric vehicles	**High (oil)**	**High**	**Medium**	**Medium**	Smart charging
Grid reinforcement	Indirect	**High**	**Medium**	**High**	Essential
Battery storage	Indirect	**Med–High**	**Medium**	**Medium**	Essential
Demand response	Indirect	**Med–High**	**High**	**Med–High**	Essential
Biomethane	**Medium**	**Medium**	**Med–Low**	**Low**	Targeted
Green hydrogen	**Sector-spec.**	**Med–High**	**Low–Med**	**Low–Med**	Targeted

Core message: The strongest options also make the whole system less vulnerable, not only those that cut imports directly.

FINAL RANKING:
A COORDINATED PORTFOLIO

1. **Energy efficiency**
   Reduces demand before the system must supply it
2. **Renewable electricity**
   Main domestic low-carbon supply-side pillar
3. **Electrification**
   Replaces imported gas and oil in heating and transport
4. **Grid flexibility**
   Makes renewables and electrification reliable & scalable
5. **Biomethane & hydrogen**
   Useful only for hard-to-electrify sectors



Final interpretation

- Efficiency shrinks the problem
- Renewables provide domestic electricity
- Electrification changes final demand
- Flexibility makes the system work
- Renewable gases cover residual difficult uses

POLICY PACKAGE: WHAT ITALY SHOULD DO

The objective is not full self-sufficiency, but lower systemic vulnerability — coordinate the portfolio, do not subsidise technologies in isolation.

Priority	Policy direction	Why it matters
1. Efficiency first	Renovate buildings, improve industrial efficiency	Reduces gas and electricity demand
2. Accelerate renewables	Faster permitting, auctions, grid-aware planning	Increases domestic low-carbon supply
3. Electrify final uses	Heat pumps and electric vehicles	Replaces gas in buildings and oil in transport
4. Invest in flexibility	Grids, storage, demand response, interconnections	Makes renewables reliable and scalable
5. Renewable gases selectively	Biomethane and hydrogen for hard-to-electrify uses	Avoids wasting scarce resources
6. Protect affordability	Targeted support, tariff reform, vulnerable consumers	Maintains social acceptance
7. Improve governance	Coordinate energy, industry, buildings & transport	Prevents bottlenecks and fragmentation

SEQUENCING: SHORT, MEDIUM & LONG TERM

A credible strategy must be sequenced across demand, supply, infrastructure and regulation.

Horizon	Main priorities	Expected contribution
Short term	Efficiency, demand response, permitting reform, better use of existing infrastructure	Fast cut in demand and price vulnerability
Medium term	Renewables deployment, heat pumps, EVs, batteries, grid upgrades	Structural reduction of gas and oil dependence
Long term	Stronger grids, interconnections, industrial hydrogen, selected biomethane	Deep system transformation and residual decarbonisation

Main principle: Do not wait for long-term technologies before acting — but do not electrify everything without preparing the power system.

CONCLUSION:
REDUCING VULNERABILITY, NOT SEEKING AUTARKY

Final answer: Italy should not aim for full self-sufficiency. The realistic goal is lower systemic vulnerability through a coordinated portfolio.



THE COORDINATED PORTFOLIO

- **Efficiency**
  reduces final energy demand
- **Renewables**
  increase domestic low-carbon electricity
- **Electrification**
  replaces gas and oil in final uses
- **Flexibility**
  makes the electricity system reliable & scalable
- **Biomethane & hydrogen**
  cover residual hard-to-electrify sectors

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THANK YOU FOR YOUR ATTENTION!!!

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